

$$h(x) = \frac{f(x)}{g(x)} = \frac{|x|+1}{[x]-1}$$

87. Option (a) is correct.

1. For $x < 0$

$$f(x) = -x + 1$$

So, $f'(x) = -1$,

So $f(x)$ is differentiable $\forall x < 0$

$\Rightarrow$ Statement 1 is correct.

2. At $x = 0.0001$

$g(x)$ is continuous every where except for integer value as $[x]$ is continuous $\forall x \in \mathbb{R}$ except integers.

So $g(x)$ is continuous at $x = 0.0001$, as 0.0001 is not an integer.

$\Rightarrow$ Statement 2 is correct.

3. $g(x) = [x] - 1$

$[x] \Rightarrow$ Always an integer

$$g'(x) = \frac{d}{dx}[x] - 0$$

$$g'(x) = 0 - 0 = 0$$

$\Rightarrow$ Statement 3 is incorrect.

$\therefore$ 1 and 2 only correct.

88. Option (a) is correct.

$$\lim_{x \rightarrow 0^-} h(x) + \lim_{x \rightarrow 0^+} h(x) = \lim_{x \rightarrow 0^-} \frac{|x|+1}{[x]-1} + \lim_{x \rightarrow 0^+} \frac{|x|+1}{[x]-1}$$

$$= \frac{0+1}{-1-1} + \frac{0+1}{0-1}$$

$$\left\{ \begin{array}{l} [0-h] = -1 \\ [0+h] = 0 \end{array} \right\}$$

$$= -\frac{1}{2} - 1$$

$$= -\frac{3}{2}$$

89. Option (d) is correct.

$$\phi(a) = \int_a^{a+100\pi} |\sin x| dx$$

As period of $|\sin x| = \pi$ and $|\sin x| = \sin x$ in the interval 0 to π .

$$\phi(a) = \int_0^{100\pi} |\sin x| dx$$

$$= 100 \int_0^\pi |\sin x| dx$$

$$= 100 \int_0^\pi \sin x dx$$

$$\phi(a) = 100(-\cos x)_0^\pi$$

$$= 100 \times 2$$

$$= 200$$

90. Option (a) is correct.

$$\phi(a) = 200$$

$$\phi'(a) = 0$$

Solution for Q.no. 91 and 92:

$$\text{Given, } y = 2f(x) + ax - b \quad \dots(i)$$

Differentiating given function w.r.t. 'x' we get,

$$\Rightarrow \frac{dy}{dx} = 2f'(x) + a(1) \quad \dots(ii)$$

Again differentiating given function, we get,

$$\Rightarrow \frac{d^2y}{dx^2} = 2f''(x) \quad \dots(iii)$$

91. Option (c) is correct.

Since, $f(x)$ has a local maximum at $x = 0$. Hence,

$$f(0) = 0 \text{ and } f'(0) < 0 \quad \dots(iv)$$

Therefore, option (c) is correct.

92. Option (b) is correct.

Since, y has a relative maxima at $x = 0$. Hence,

$$\left(\frac{dy}{dx} \right)_{x=0} = 0$$

$$\Rightarrow 2f'(0) + a = 0 \quad [\text{from (ii)}]$$

$$\Rightarrow 2(0) + a = 0 \quad [\text{from (iv)}]$$

$$\Rightarrow a = 0$$

$$\text{also } \left(\frac{d^2y}{dx^2} \right)_{x=0} < 0$$

$$\Rightarrow 2f''(0) < 0$$

$$\Rightarrow f''(0) < 0$$

Clearly y has a relative maxima for $a = 0$ and all values of 'b'.

Solution for Q.no. 93 and 94:

$$\text{Given, } f(x) = |x-1| \quad \dots(i)$$

$$g(x) = [x] \quad \dots(ii)$$

$$\text{and } h(x) = f(x) \cdot g(x)$$

$$\Rightarrow h(x) = |x-1| \cdot [x] \quad \dots(iii)$$

93. Option (a) is correct.

$$\int_{-1}^0 h(x) dx = \int_{-1}^0 f(x) \cdot g(x) dx$$

$$= \int_{-1}^0 |x-1| \cdot [x] dx$$

$$= \int_{-1}^0 (1-x) \cdot (-1) dx$$

$$= \int_{-1}^0 (x-1) dx$$

$$= \left[\frac{x^2}{2} - x \right]_{-1}^0 = \left[0 - \left(\frac{1}{2} - 1 \right) \right]$$

$$= -\frac{3}{2}$$